



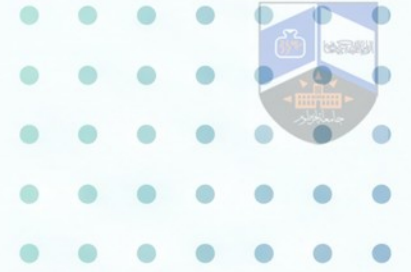
UNIVERSITY OF KHARTOUM · FACULTY OF ENGINEERING · ELECTRICAL & ELECTRONIC ENGINEERING

Improving Arabic RAG Retrieval Quality with LLM-based Query Enhancement

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AGENDA

01

Introduction



04

Methodology



02

**Problem &
Research Question**



05

**Results &
Discussion**



03

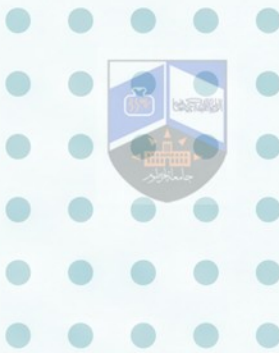
Objectives



06

**Conclusion &
Future Work**





INTRODUCTION • A LANGUAGE MODEL IS FROZEN IN TIME

Parametric memory

Frozen at training time

Retraining is expensive

Hallucination



Everything it knows is sealed inside.

? A question it was never trained on

ما الشروط التي تسمح للطلاب بتأجيل دراسته
في جامعة الخرموم؟

“

×

It does not say “I don’t know.” It answers —
fluently, confidently, and wrongly.

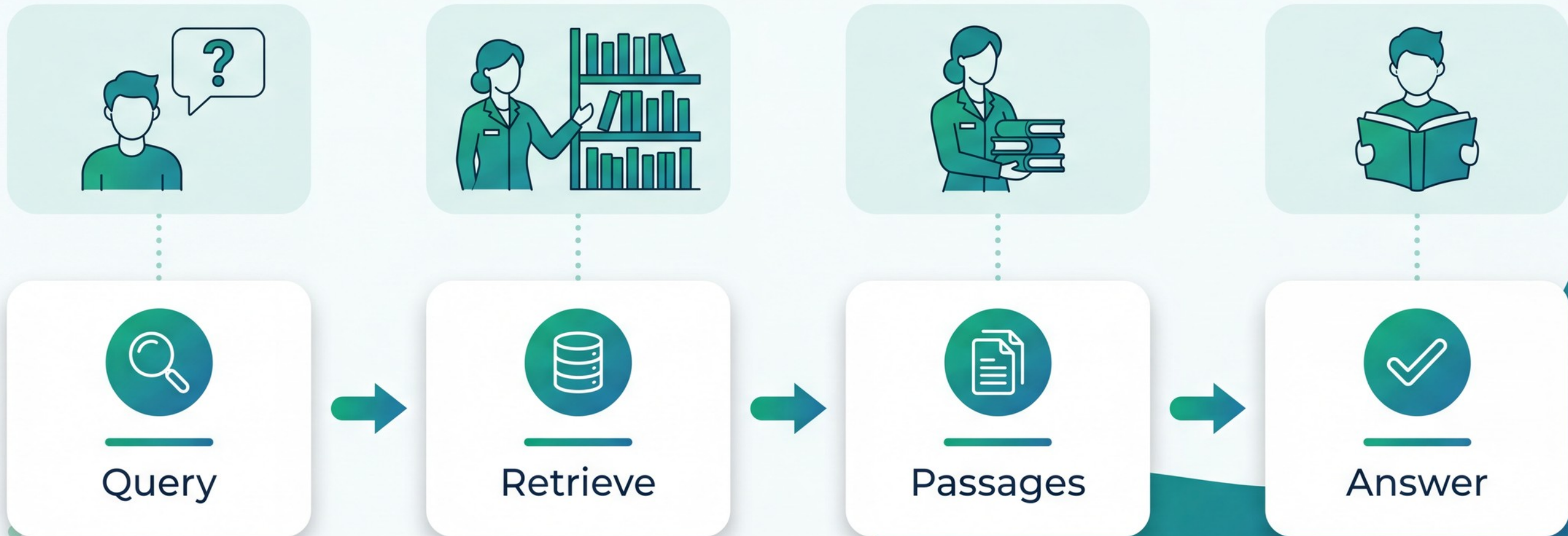


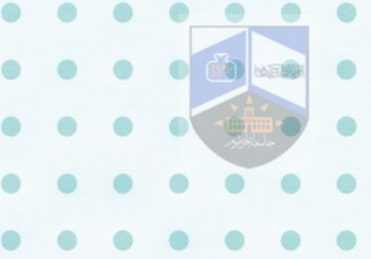
INTRODUCTION •

DON'T RETRAIN THE MODEL. GIVE IT THE DOCUMENTS.

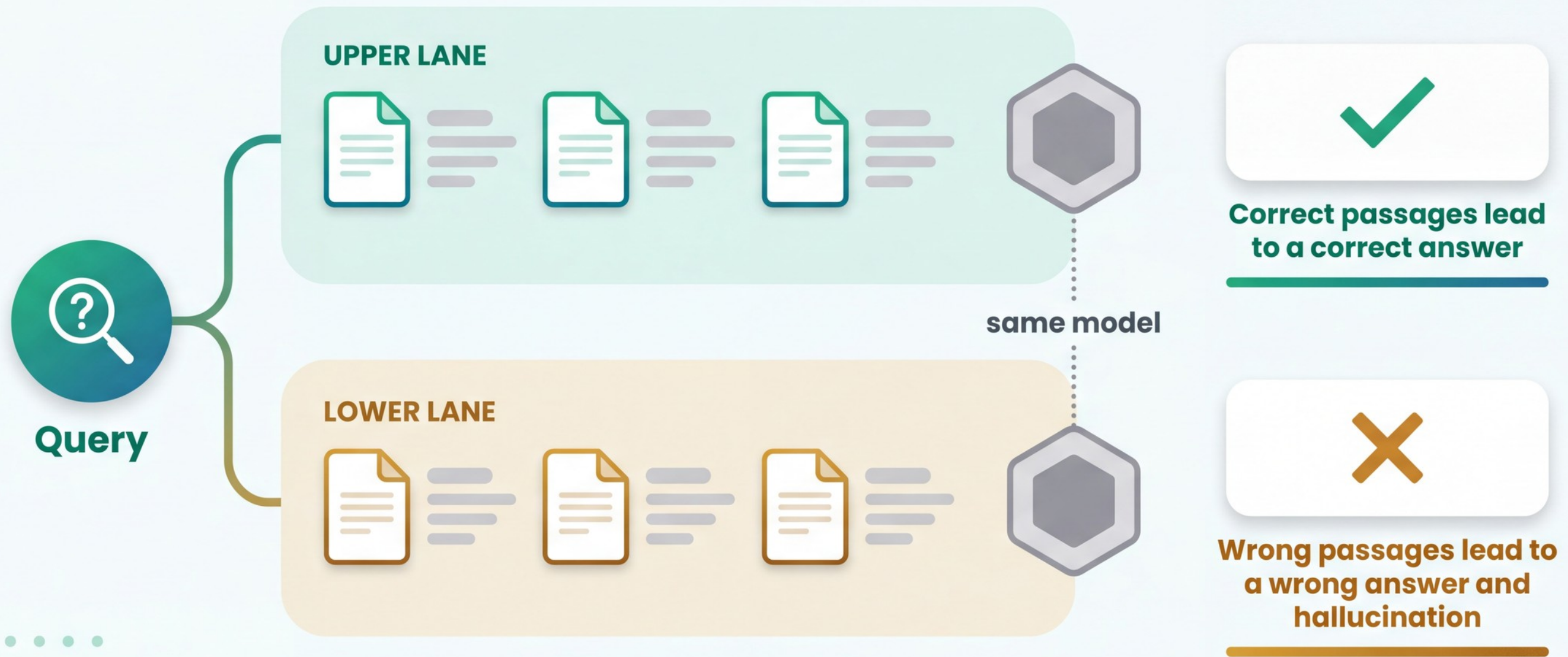
RAG

Retrieval-Augmented Generation





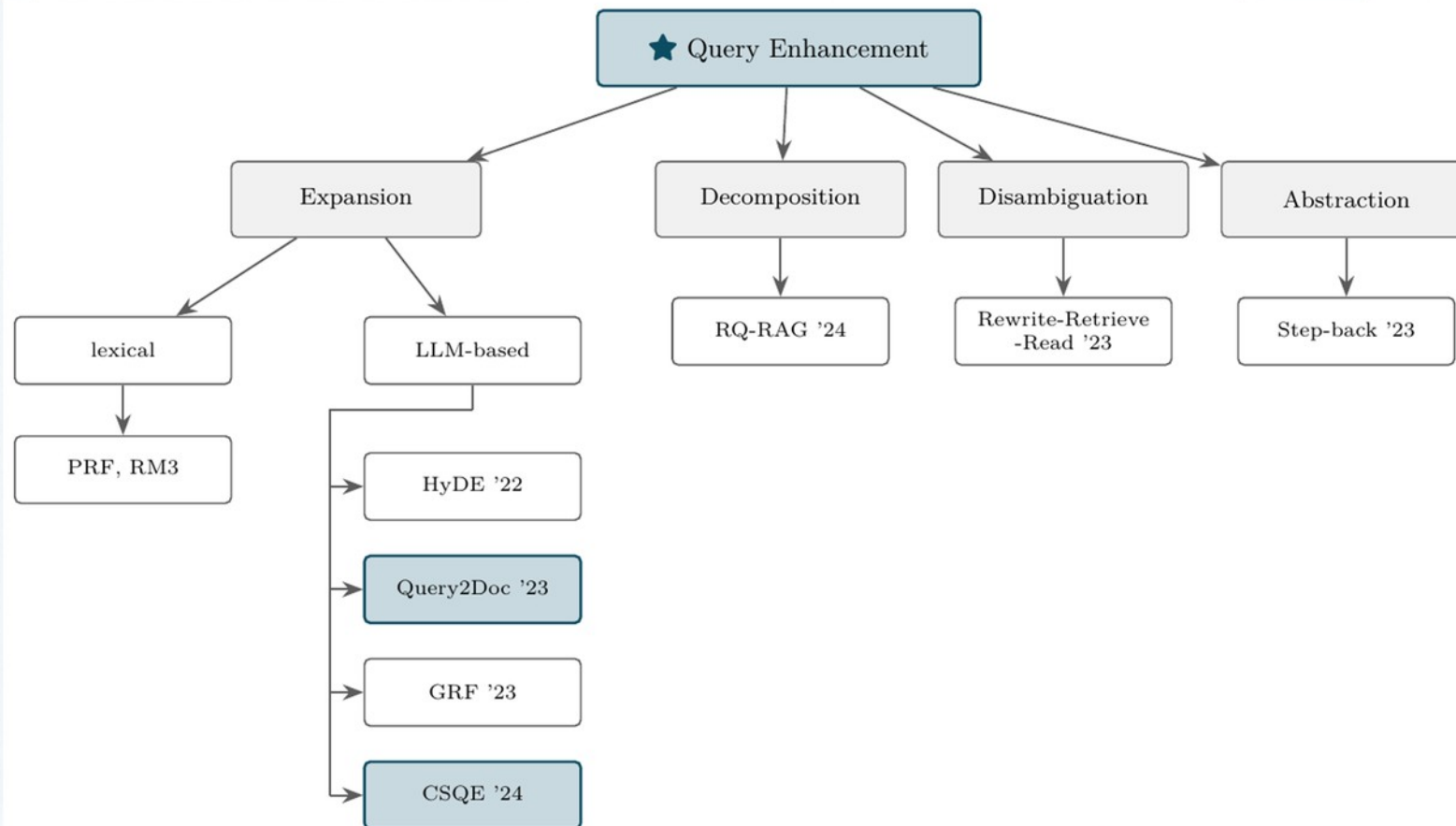
THE PROBLEM • ANSWER QUALITY CAN NEVER EXCEED RETRIEVAL QUALITY





THE GAP ● RICH IN ENGLISH. ALMOST EMPTY IN ARABIC.

Query Enhancement — repair the query before it reaches the retriever



English

Dozens of published techniques



Arabic

Almost nothing for Arabic

THE GAP • BUT ARABIC IS NOT ENGLISH



MORPHOLOGY

كتب • كاتب •
مكتوب • كتاب

One root, dozens of
surface forms



ORTHOGRAPHY

مكتبة / مكتبه
إسلام / اسلام

Same word, two
accepted spellings



DIACRITICS

كُتِبَ / كَتَبَ
عِلْمٌ / عَلَّمَ

Same letters,
different word



DIGLOSSIA

عايز أعرف شروط التأجيل

ما هي شروط تأجيل الدراسة

People ask in **dialect**,
knowledge is written
in **MSA**

RESEARCH QUESTION

“

To what extent can LLM-based query enhancement improve Arabic information retrieval, across sparse, dense and hybrid retrieval, using openly available models between 2 and 8 billion parameters?



Beyond validation – how do different retrievers interact with query enhancement?





OBJECTIVES

01

Establish dense and sparse baselines and measure their complementarity

02

Run a systematic error analysis to identify the failure modes

03

Adapt Query2Doc for Arabic zero-shot on free cloud GPUs

04

Compare ten open LLMs from 2 to 8 billion parameters

05

Analyse how expansions interact with sparse versus dense retrieval

06

Investigate query repetition as a remedy for term dilution

07

Establish a hybrid fusion baseline without query enhancement

08

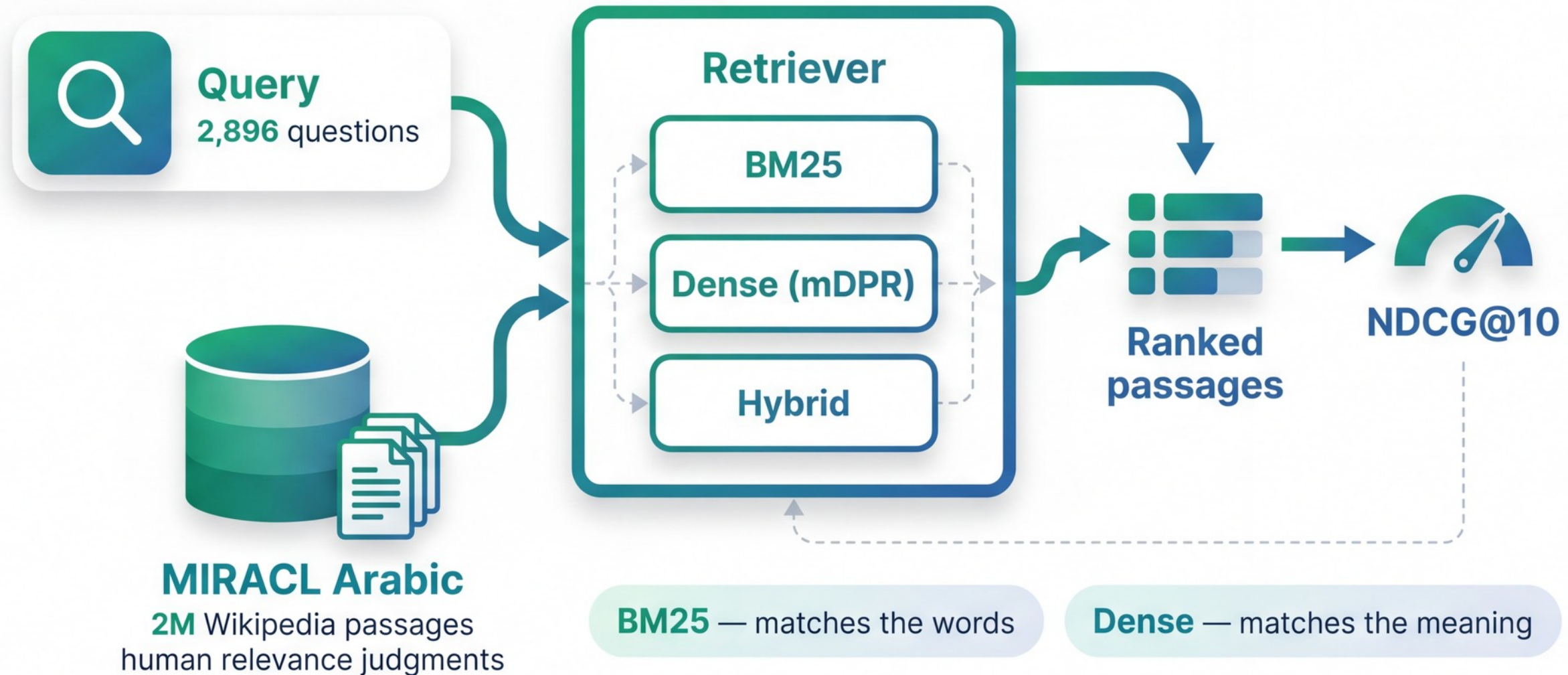
Adapt and evaluate CSQE for Arabic with component ablation

09

Determine the optimal placement of expansion in the hybrid pipeline

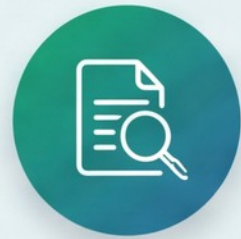


METHODOLOGY • THE SYSTEM WE BUILT





METHODOLOGY • BASELINE RESULTS, AND WHAT THEY POINTED AT



We **reproduced** the published MIRACL baselines before changing anything

0.4993

mDPR — dense baseline

0.4621

BM25S — sparse baseline

34%

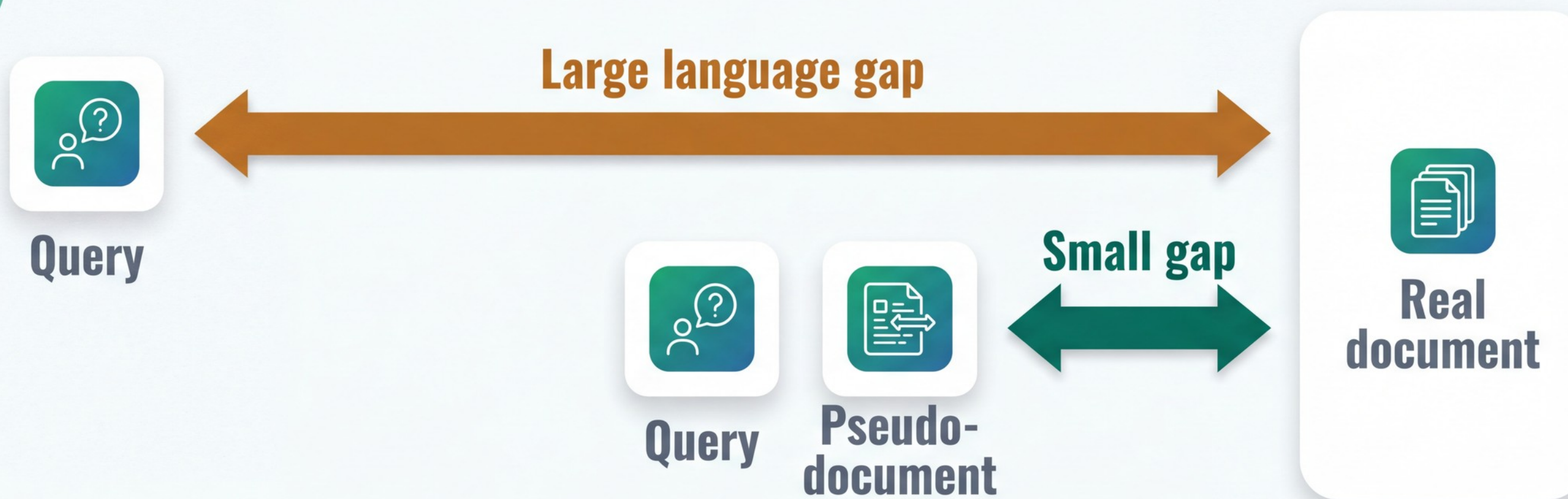
of queries failed outright

0.345

Short queries — the weakest group

Information poverty: too few words to separate one passage from two million

METHODOLOGY • QUERY2DOC - HALLUCINATION AS A FEATURE

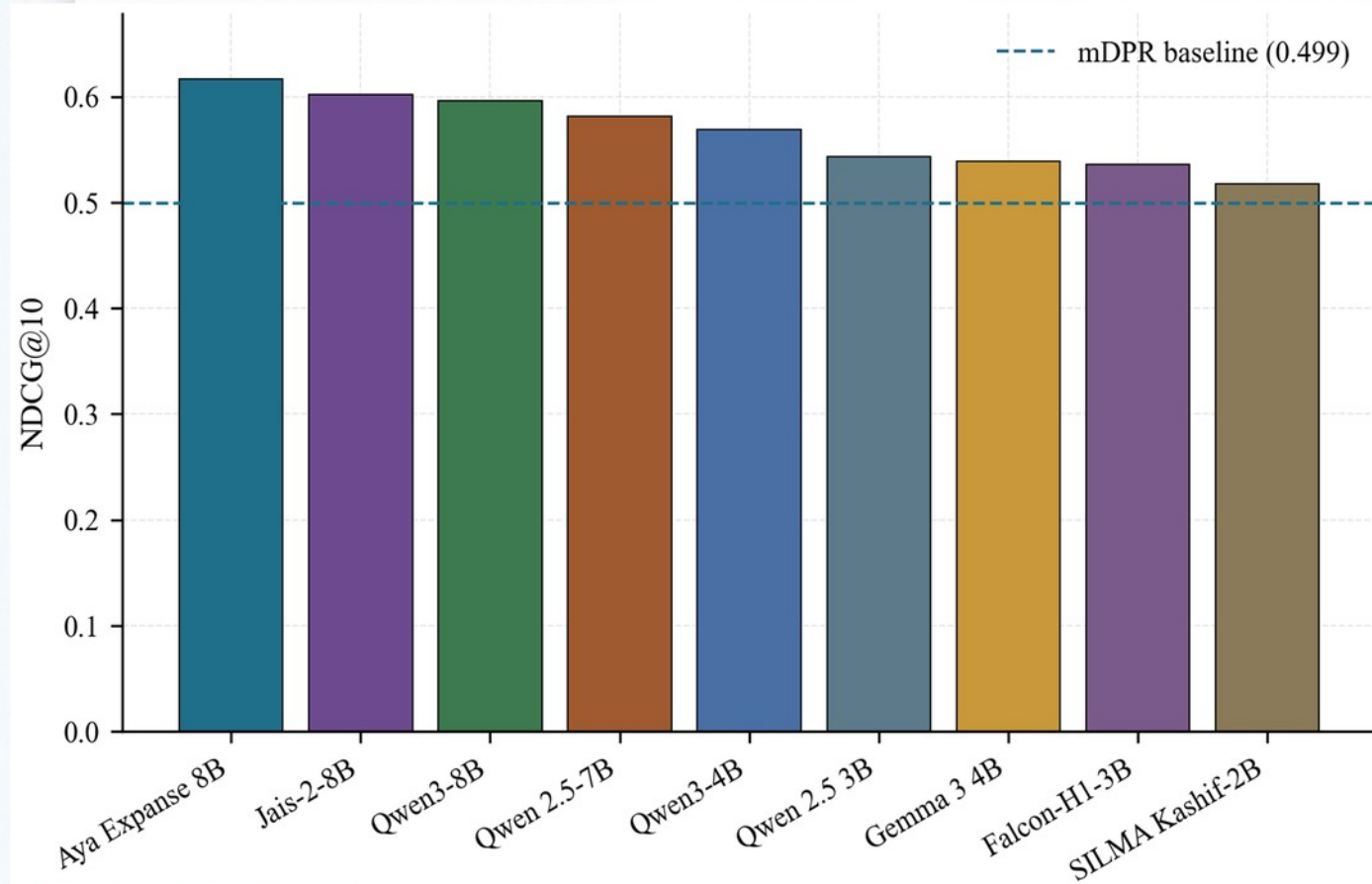


The model's guess may be wrong, but its wording matches the corpus

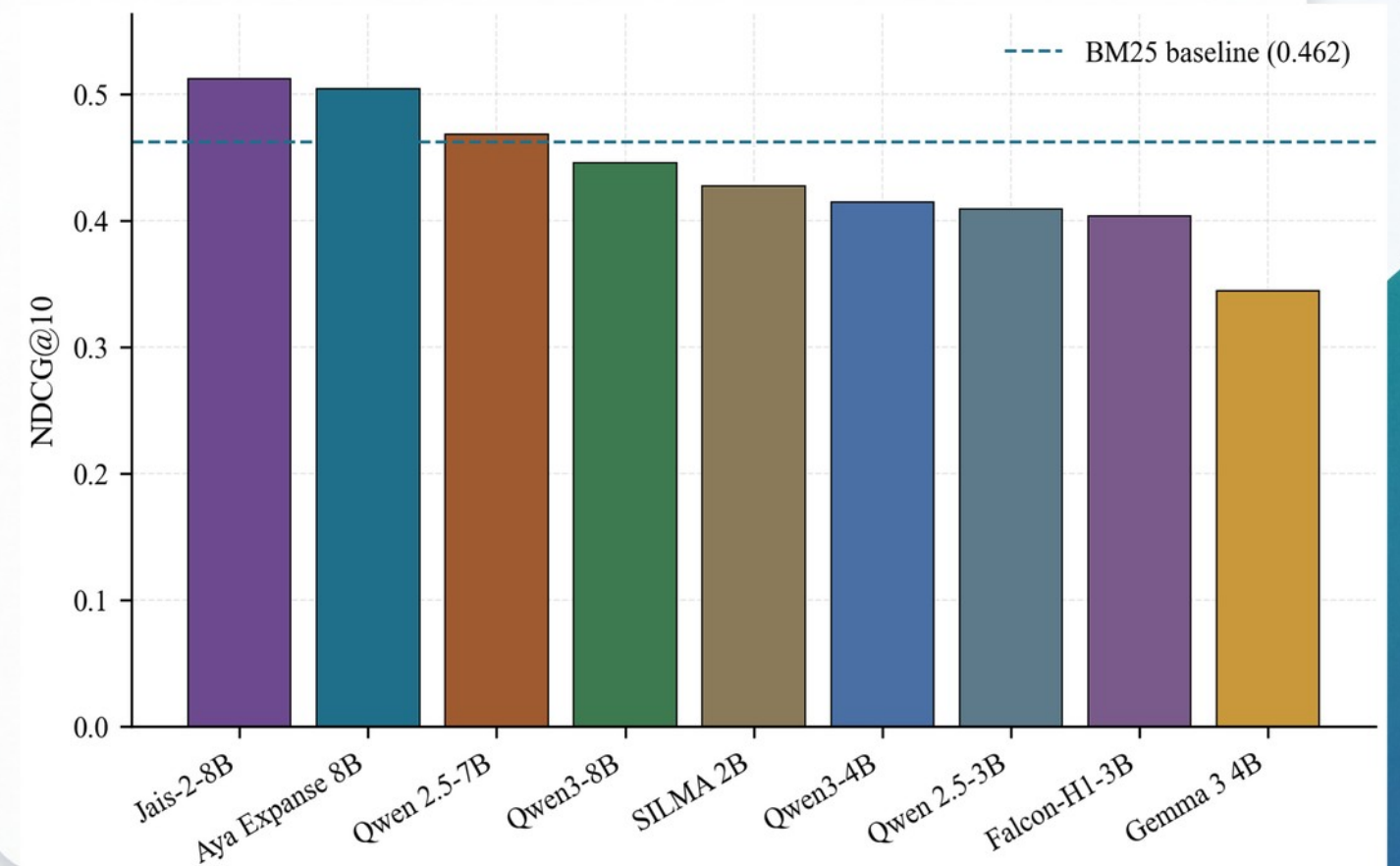


RESULTS • SAME EXPANSION. OPPOSITE EFFECT.

Dense — every model improved



BM25 — six of nine got worse





RESULTS • QUERY REPETITION RESTORES THE ORIGINAL QUERY WEIGHT

01

Expansion diluted the original query

02

We repeat the query inside it

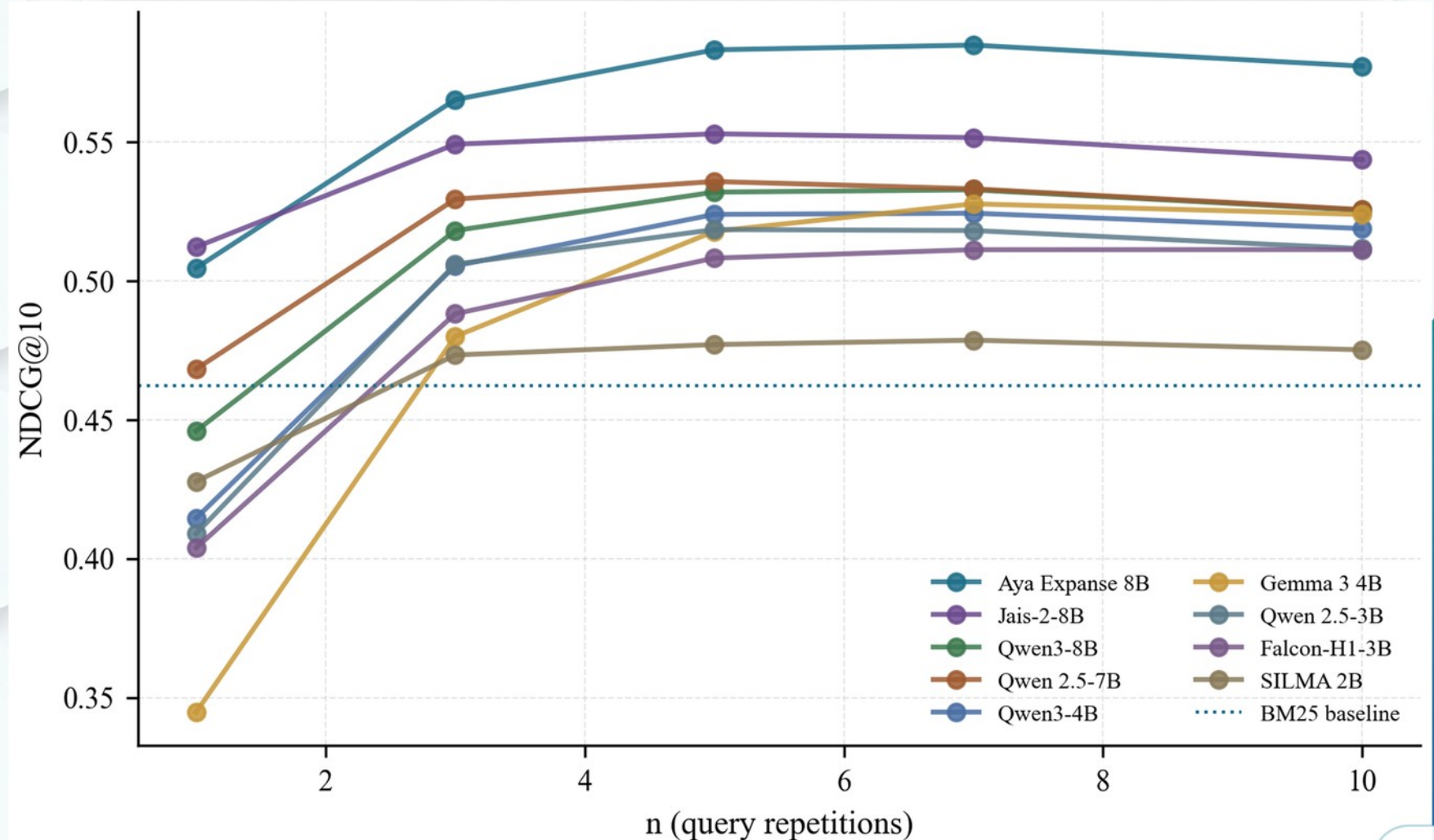
03

The optimum differs per model

0.5855

all nine models above baseline

Repetition sweep across nine models



THE CATCH • WHERE EXPANSION BREAKS



ما هي الأسماء الخمسة في اللغة العربية؟

A well-known rule of Arabic grammar



The correct answer

أب ، أخ ، حم ، فو ، ذو



What the model generated

محمد ، آدم ، إبراهيم ، إسماعيل

It listed people's names

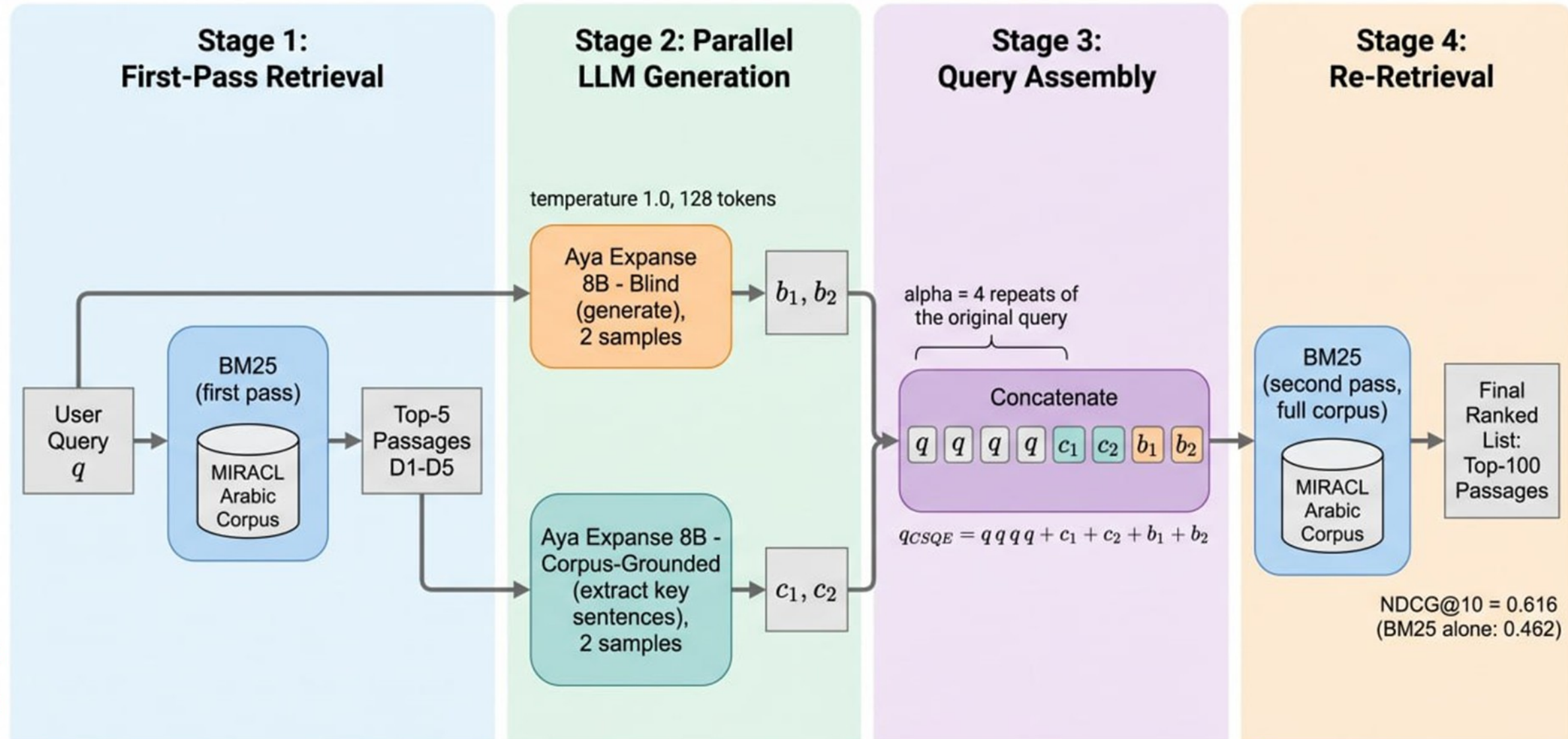
1.000
bare query



0.000
query + expansion



OUR METHOD • CORPUS-STEERED QUERY EXPANSION





THE CONTRIBUTION ●

WHICH RETRIEVER SHOULD GET THE EXPANDED QUERY?

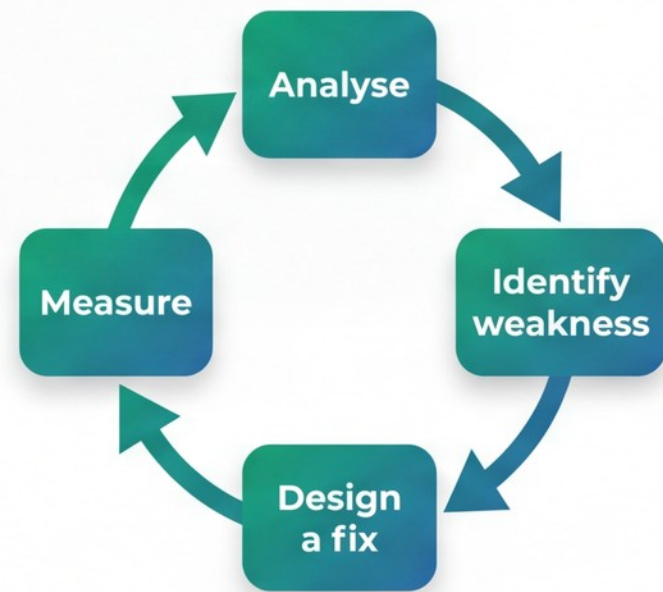
No expansion - hybrid only		0.6267
Expansion on the dense retriever only		0.6474
Expansion on both retrievers		0.6936
Expansion on BM25 only		0.7137



Fusion only pays while the two retrievers stay different. Expand **both** and they converge.



RESULTS • THE FULL JOURNEY



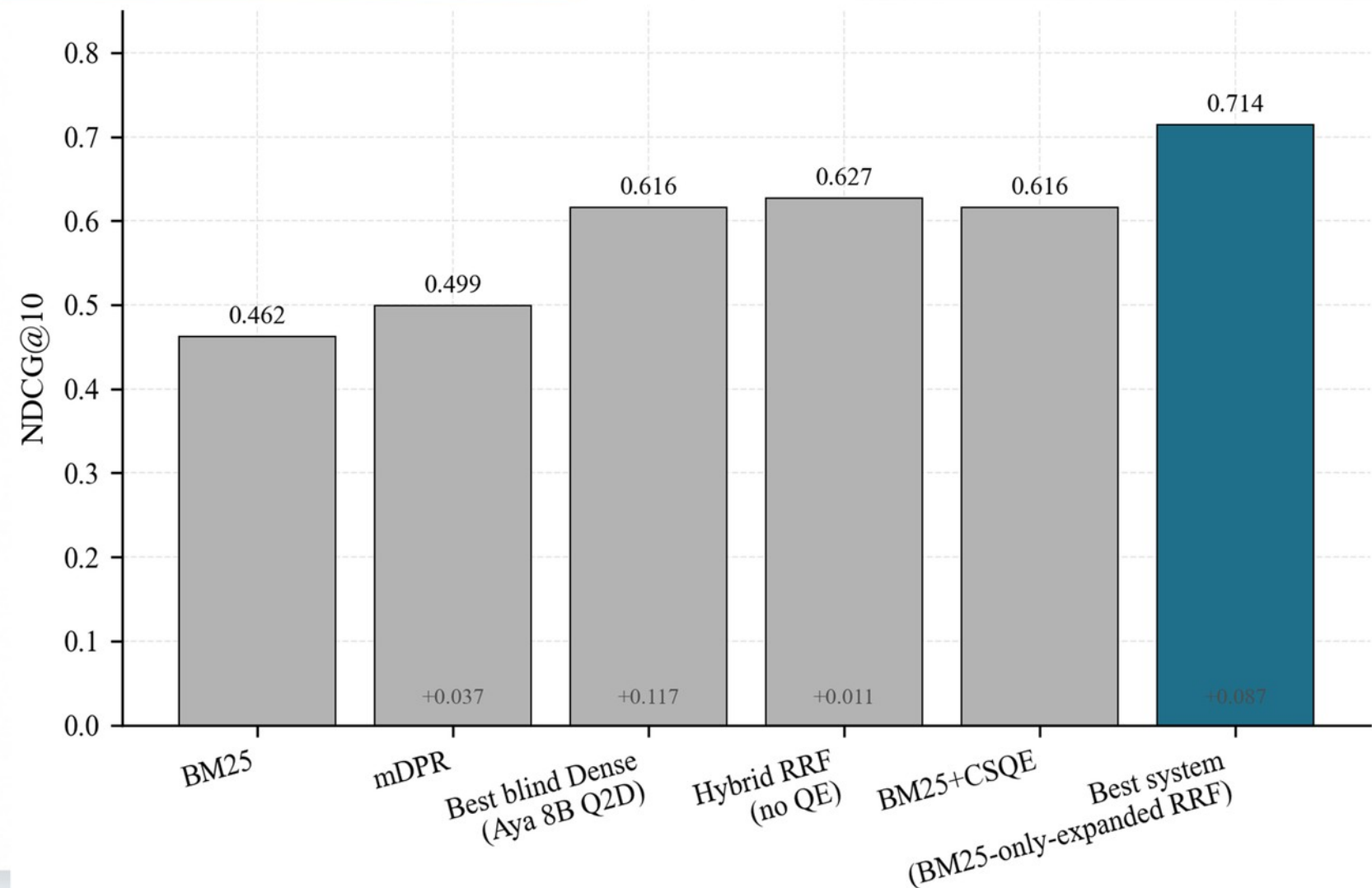
Our method, repeated

0.7137 final system

+54.5% over BM25

+13.9% over no-QE hybrid

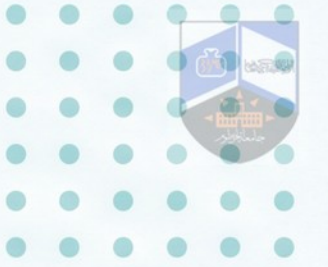
Baseline to final system





RESULTS • THE SAME TREND ACROSS ALL FOUR METRICS

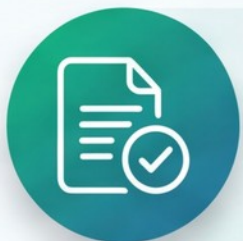
Method	NDCG@10	Recall@10	Recall@100	MRR
BM25S alone	0.4621	0.5964	0.8577	0.4836
mDPR alone	0.4993	0.6156	0.8407	0.5328
Blind QE → Dense	0.6164	0.7256	0.9001	0.6493
Blind QE + repetition → BM25	0.5855	0.7128	0.9300	0.6165
Hybrid RRF, no QE	0.6267	0.7597	0.9466	0.6517
CSQE → BM25	0.6157	0.7447	0.9422	0.6380
CSQE → Dense	0.5915	0.7073	0.8816	0.6225
★ CSQE on BM25 + Dense raw	0.7137	0.8363	0.9734	0.7362



CONCLUSIONS

Query enhancement transfers to Arabic — and the mechanisms we found are not Arabic-specific

- 01** Baselines established and error-analysed
- 02** Query2Doc transfers to Arabic zero-shot
- 03** Ten open LLMs compared under one protocol
- 04** Dense and sparse respond differently to expansion
- 05** Query repetition resolves sparse degradation
- 06** Corpus grounding validated
- 07** Retriever-specific query representation is critical



First systematic validation of LLM-based query enhancement for Arabic retrieval — ten open models, free hardware, fully reproducible. A paper is in preparation.

FUTURE WORK



DIALECTAL ARABIC

Evaluate query enhancement on **regional Arabic varieties** beyond **Modern Standard Arabic**.



FIRST-PASS QUALITY GATE

Fall back to **blind expansion** when the **first retrieval pass** is **poor**.



STRONGER EMBEDDING MODELS

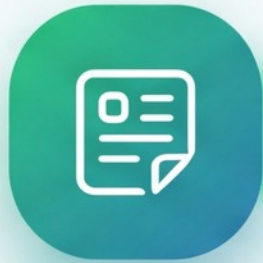
Test whether the gains hold over **more capable Arabic retrievers**.





Thank you

Questions



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